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(54) **MODULAR PUMP/MOTOR ASSEMBLY**

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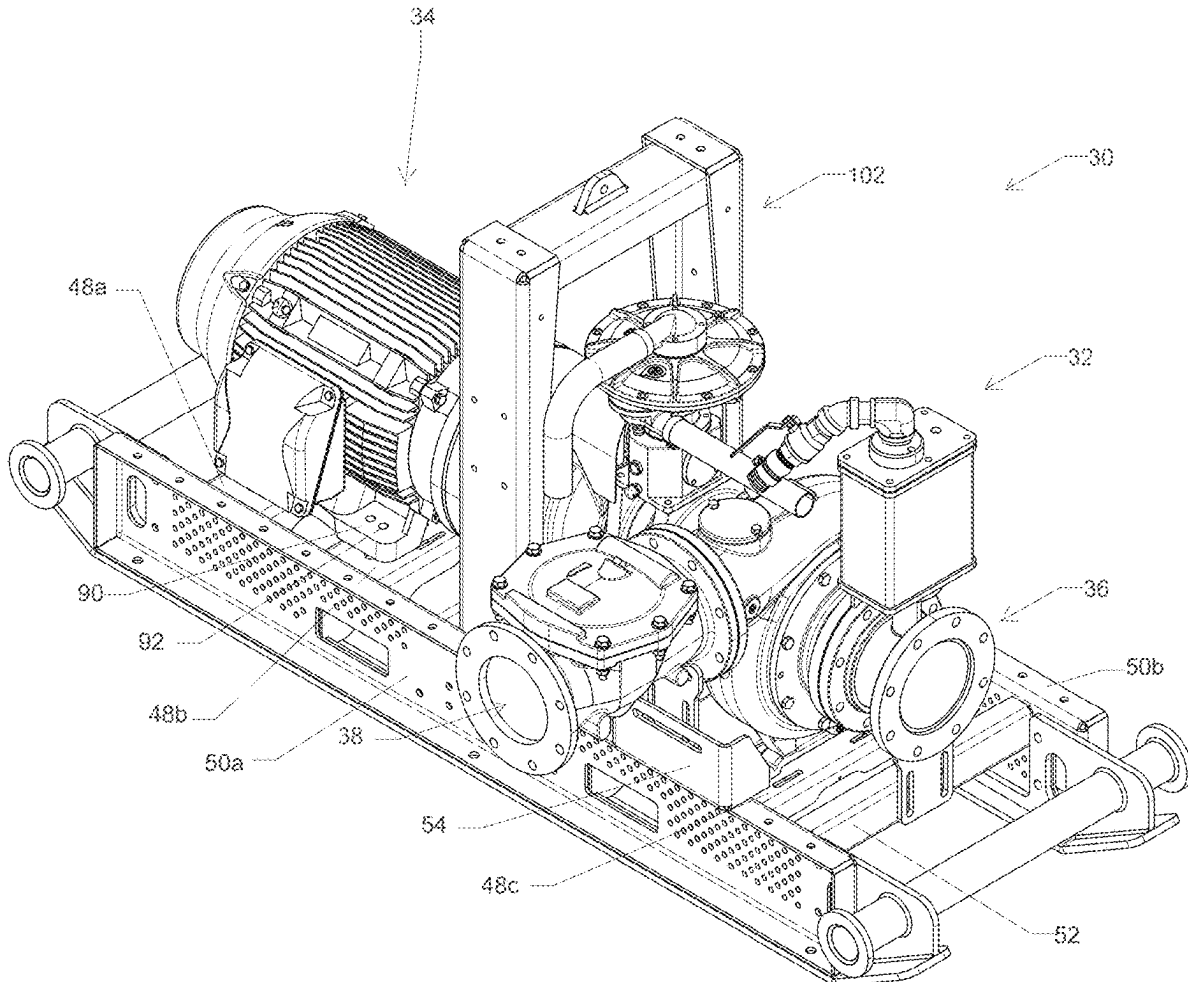
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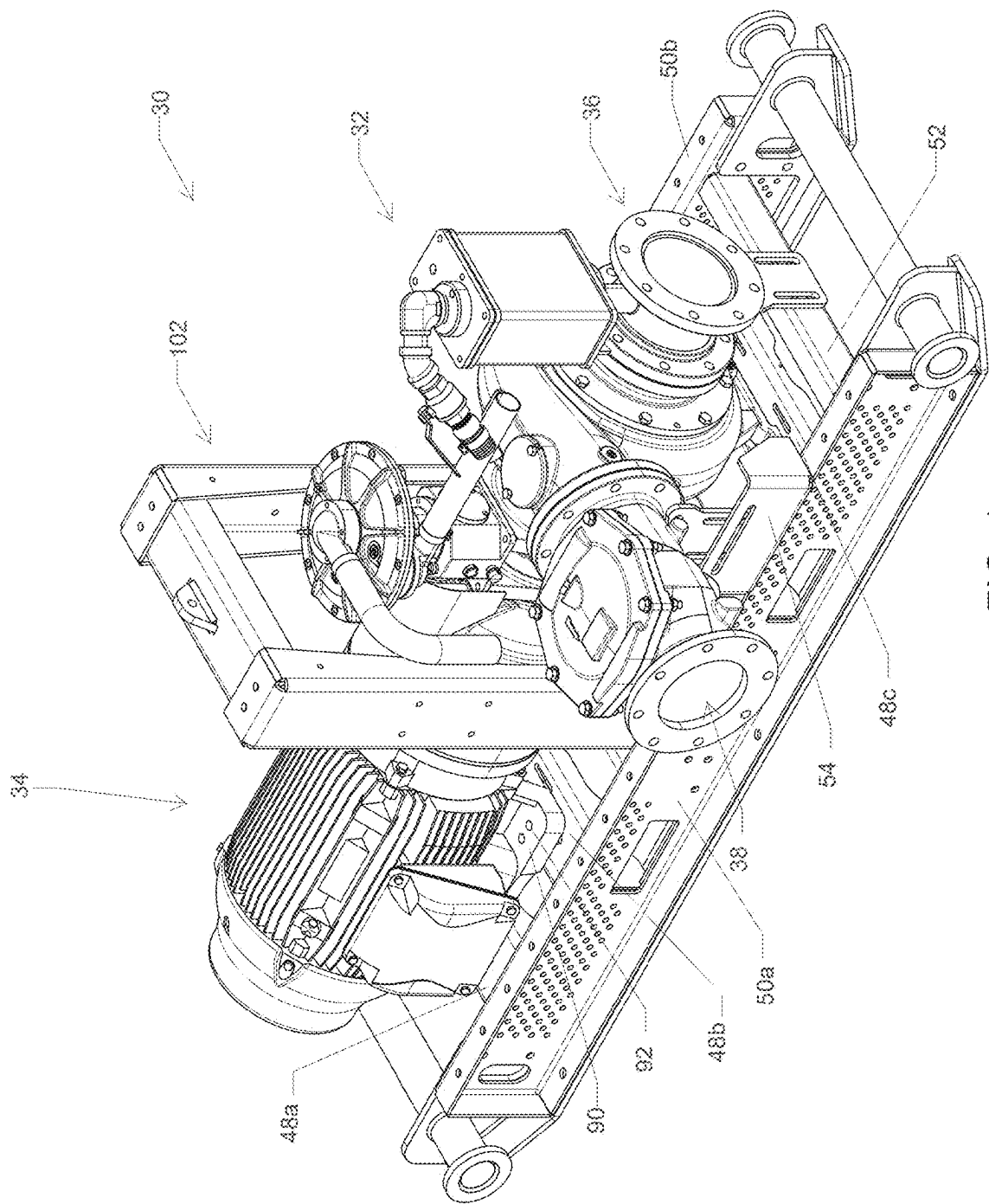
Related U.S. Application Data

(63) Continuation of application No. 18/197,986, filed on May 16, 2023.

(57) **ABSTRACT**

A modular pump and motor system including a modular frame and motor stool is disclosed.





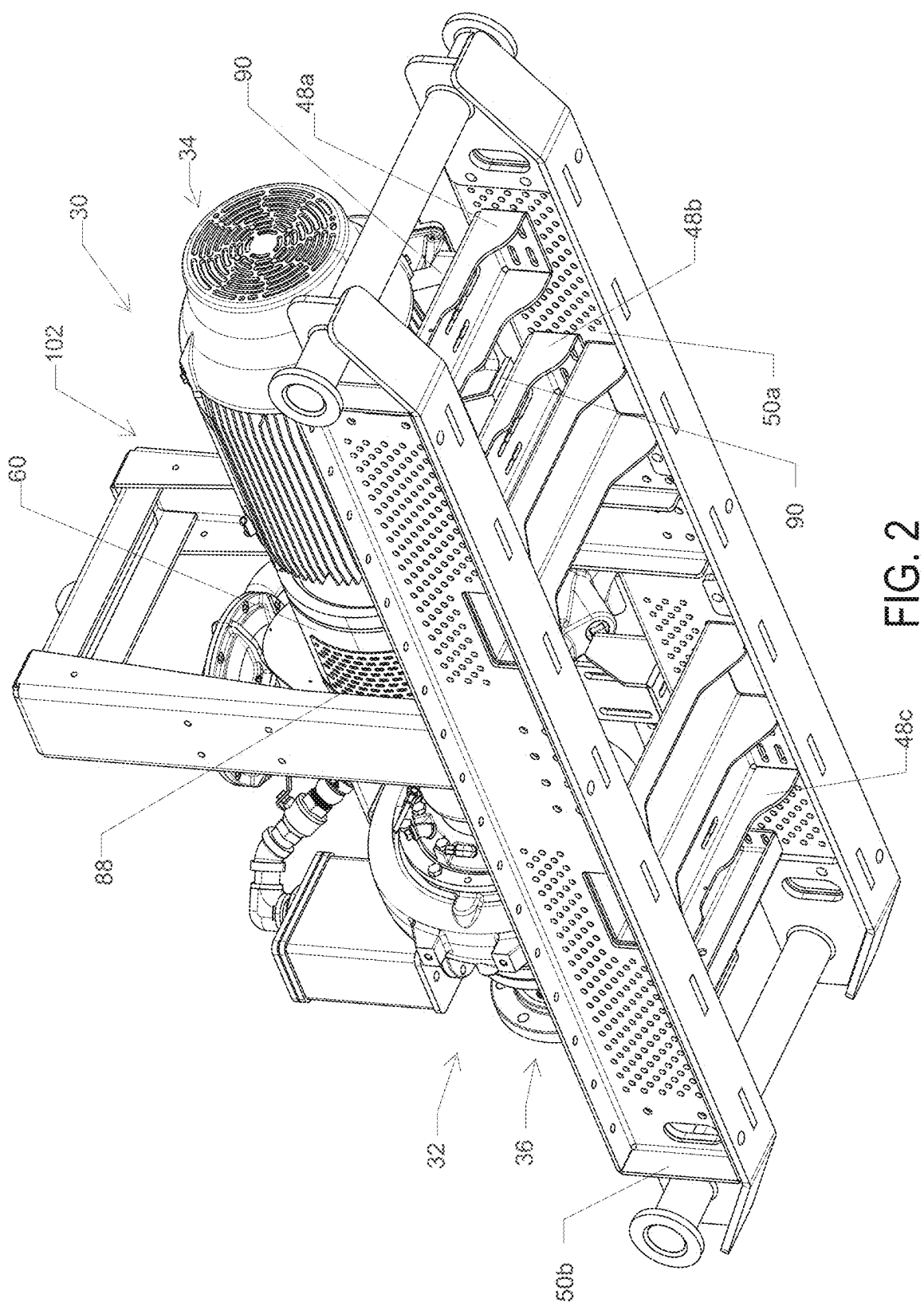


FIG. 2

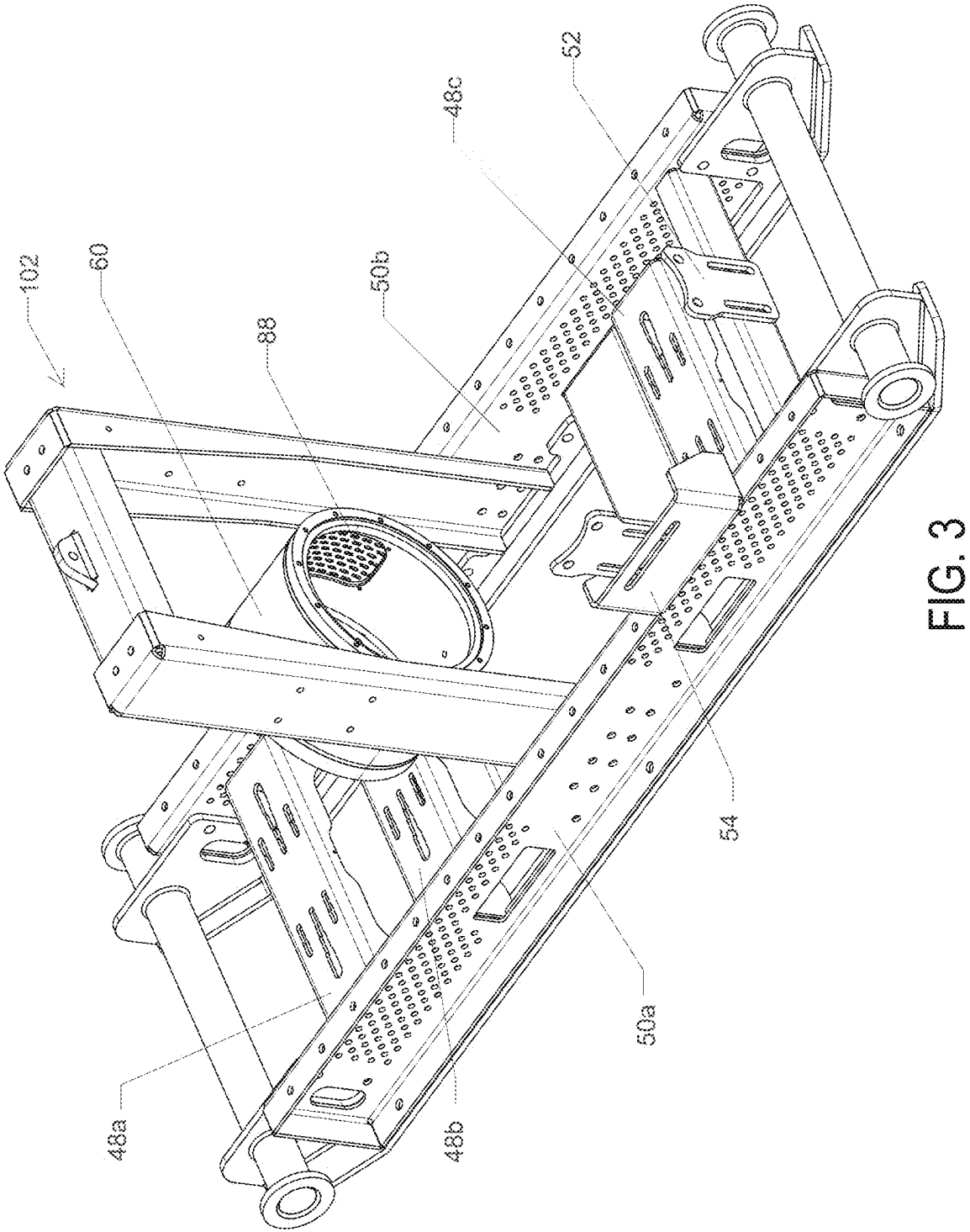


FIG. 3

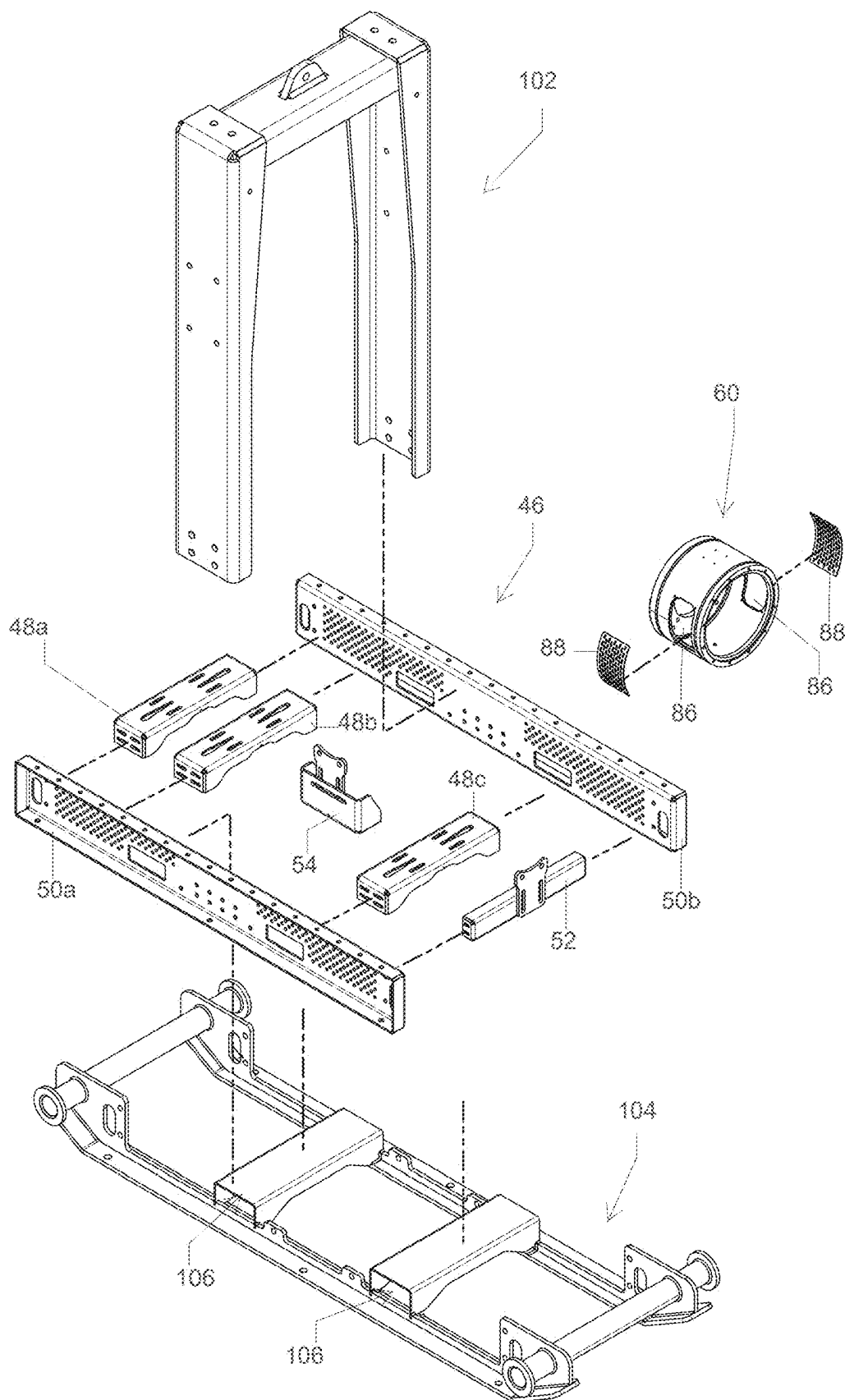


FIG. 4

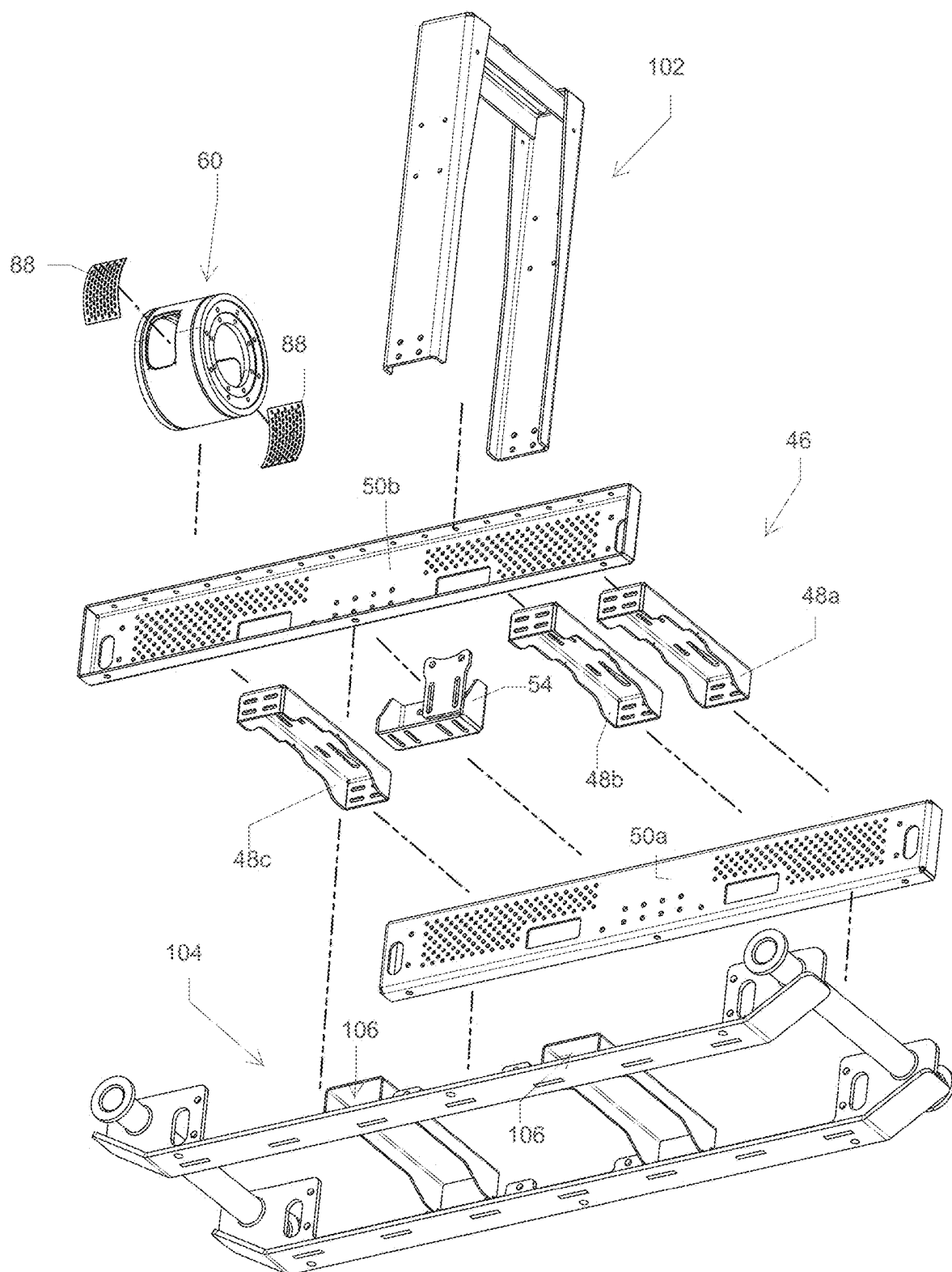
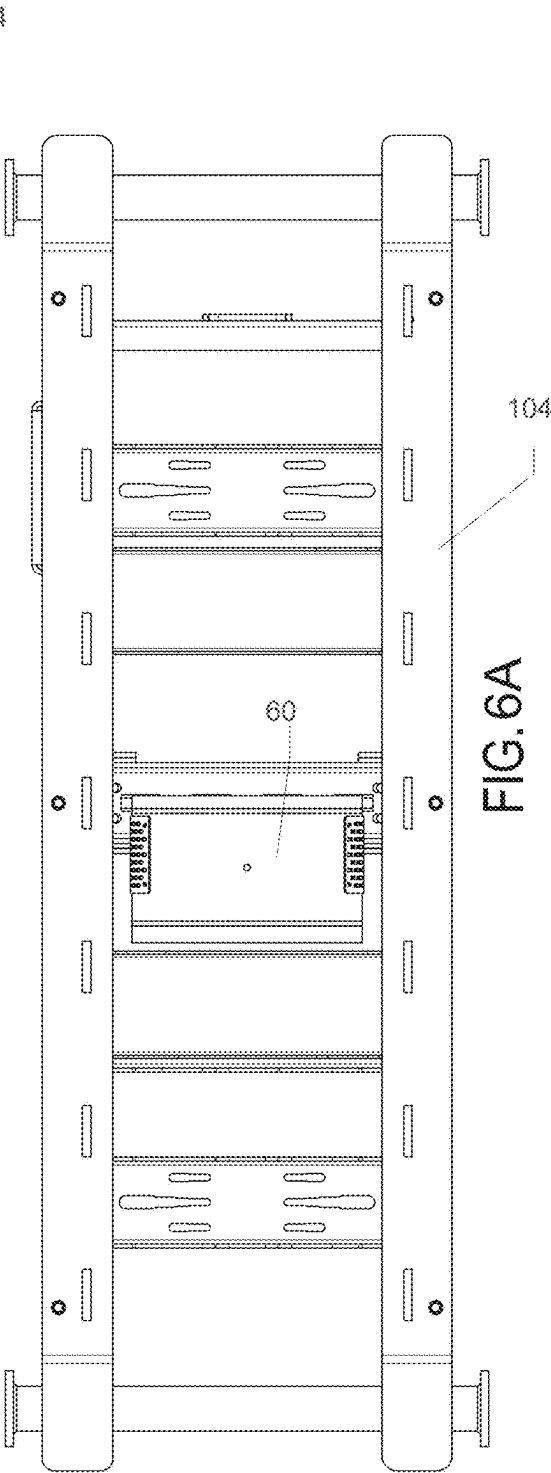
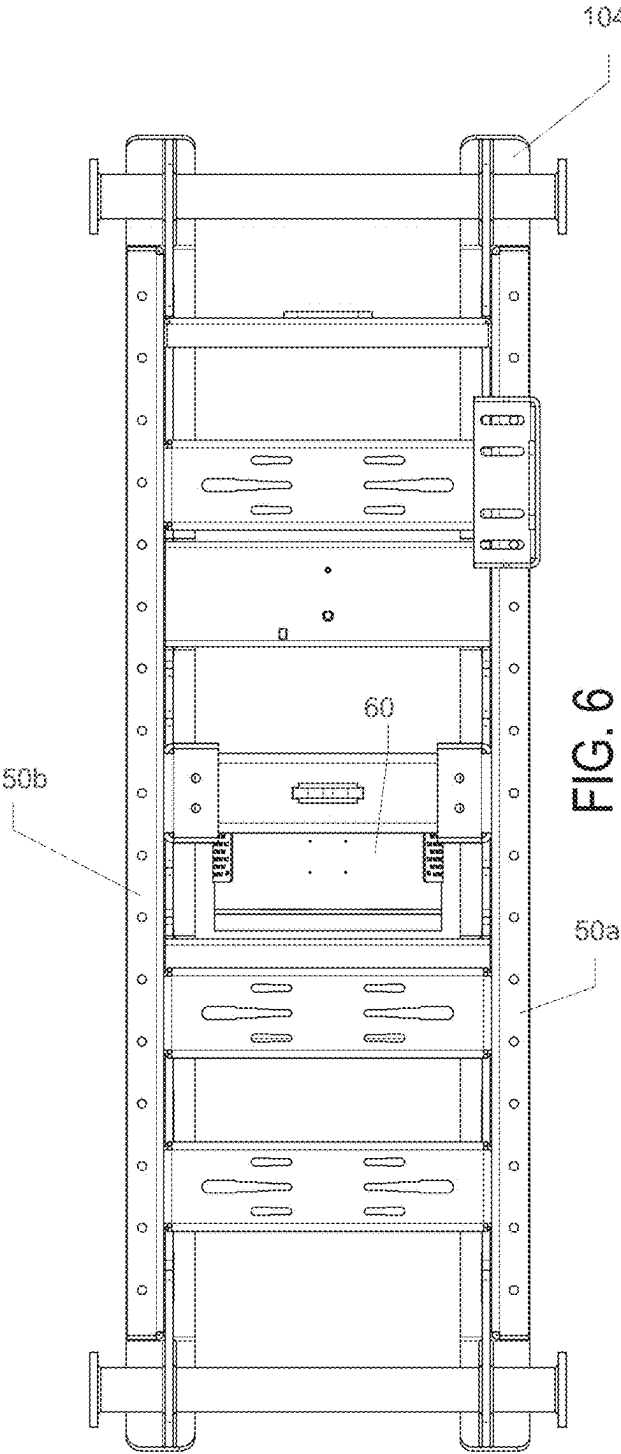


FIG. 5



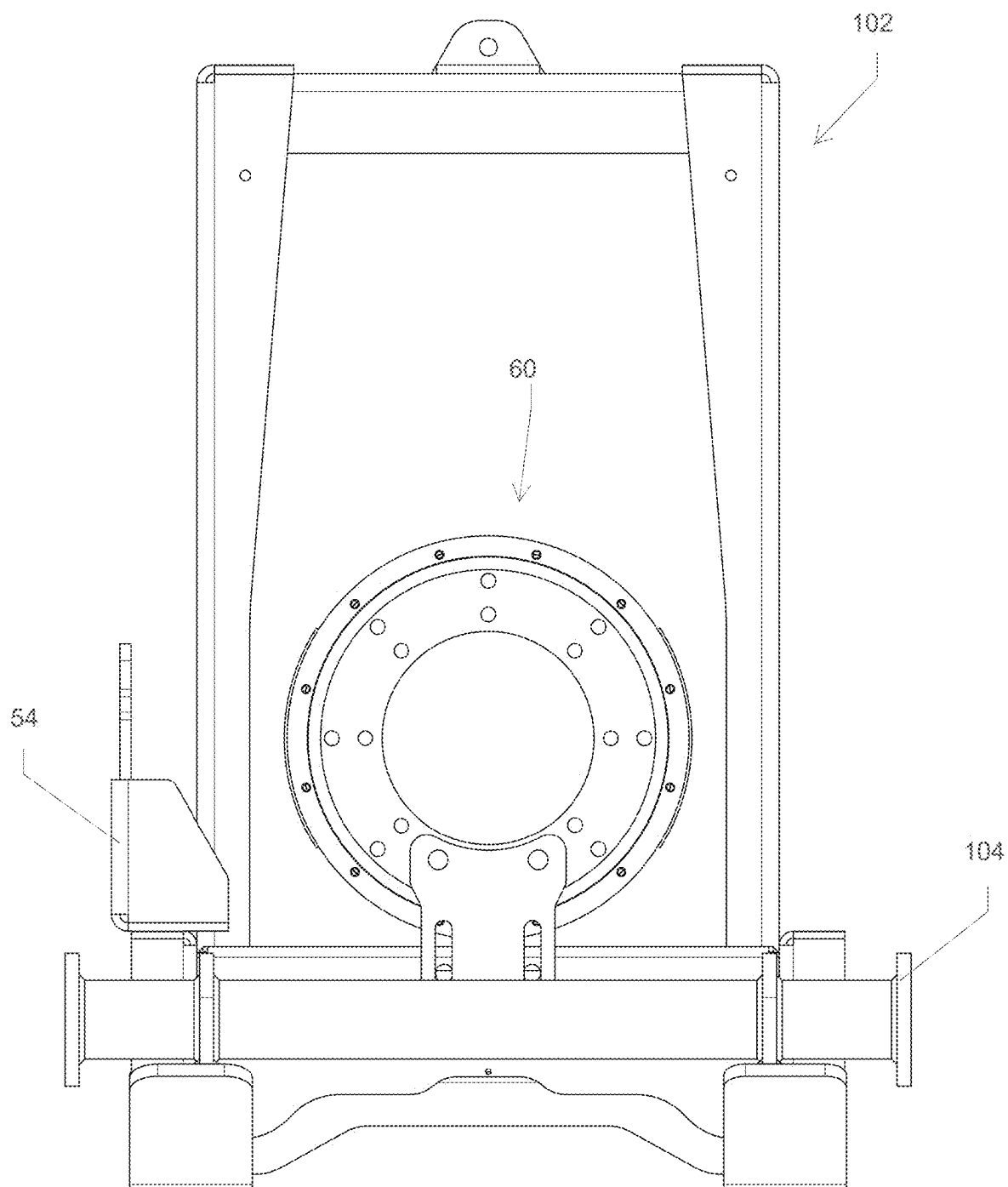


FIG. 7

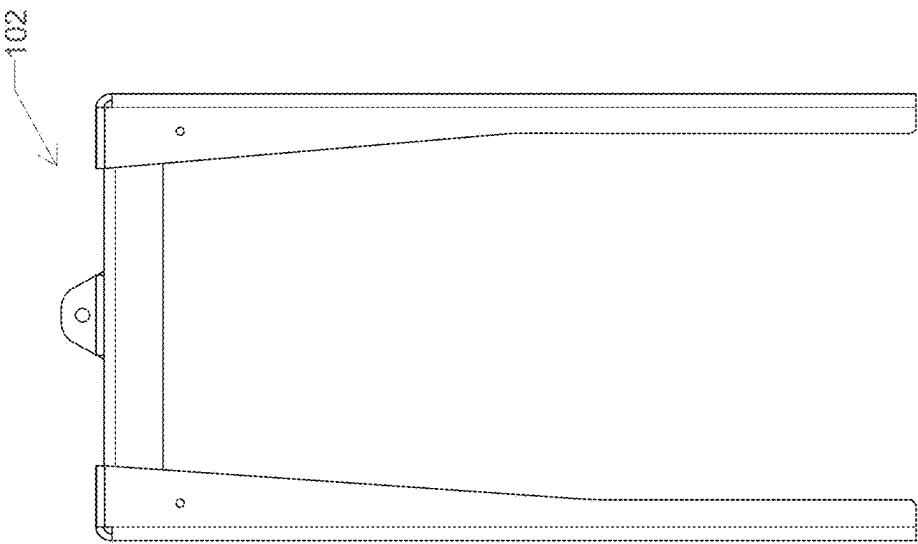


FIG. 10

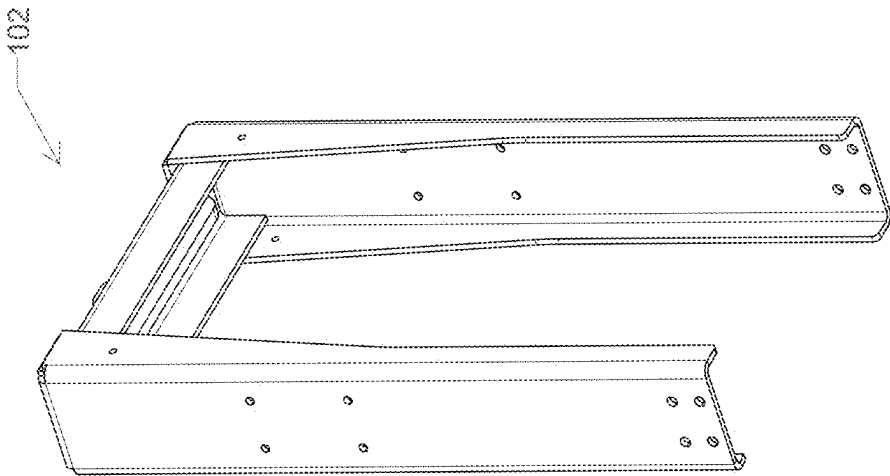


FIG. 9

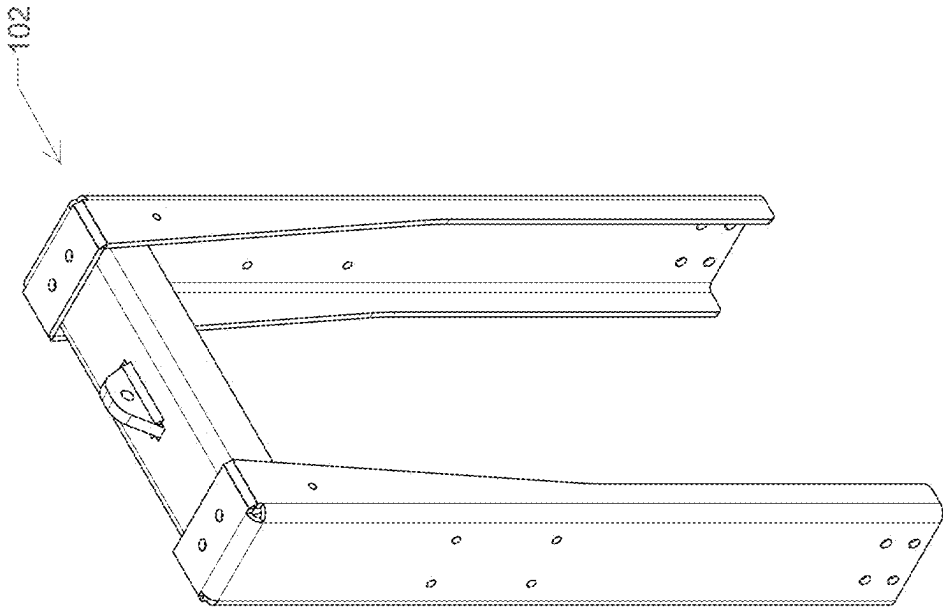


FIG. 8

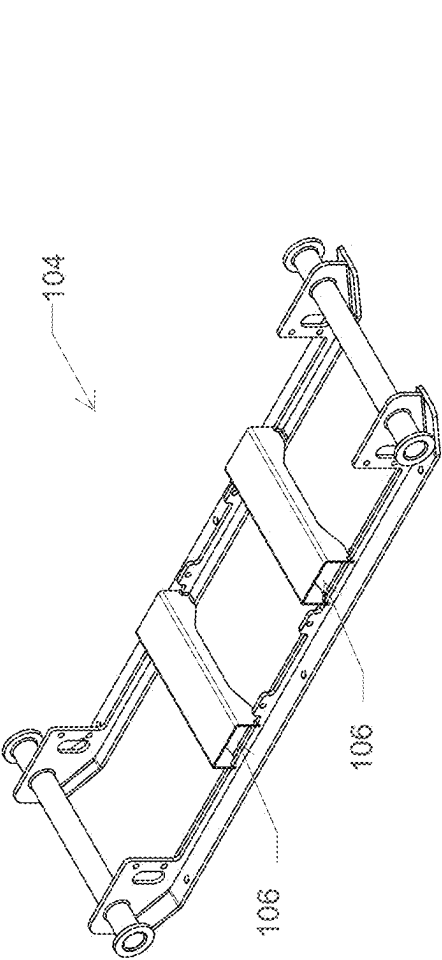


FIG. 11

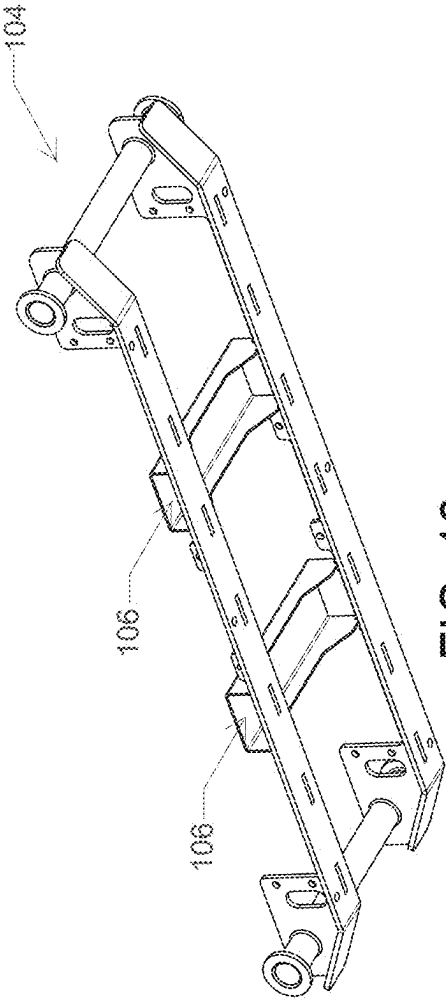
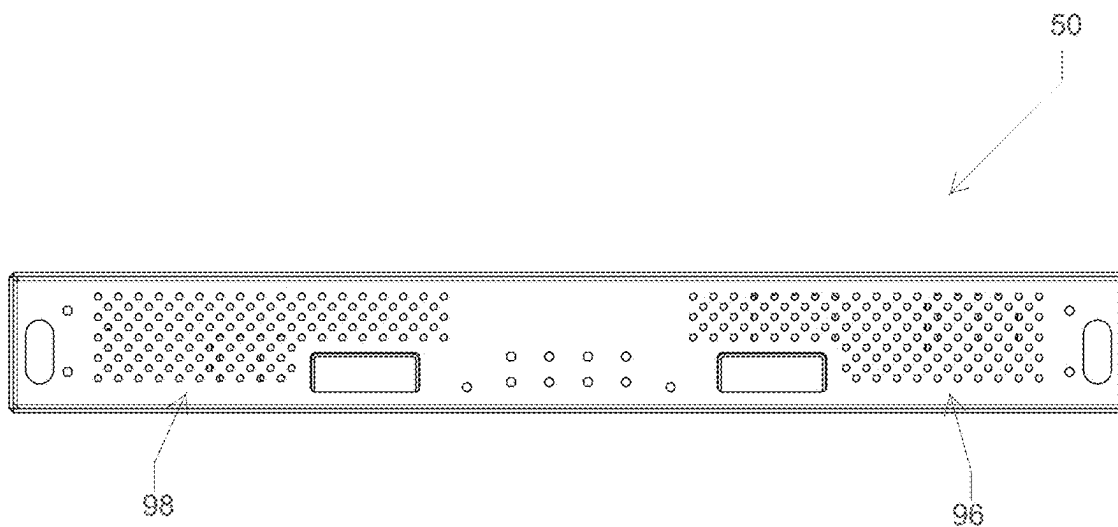
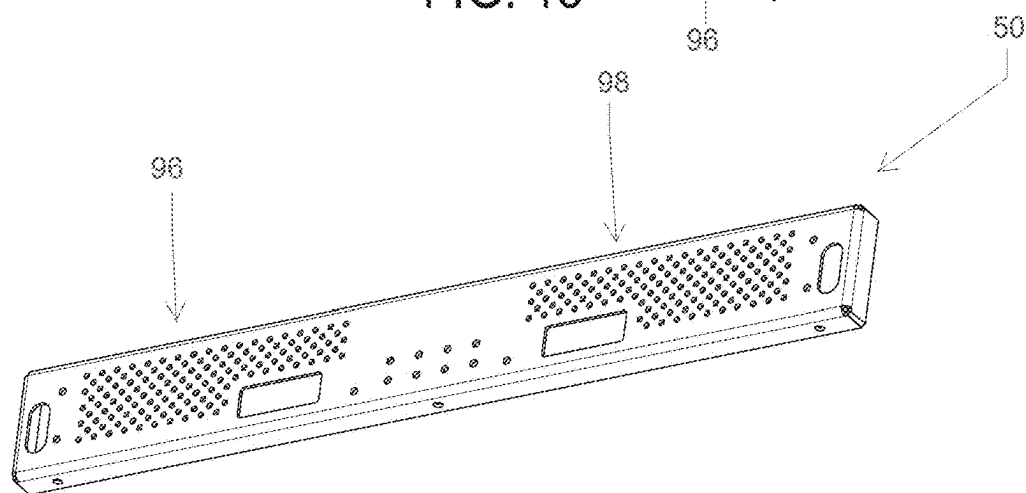
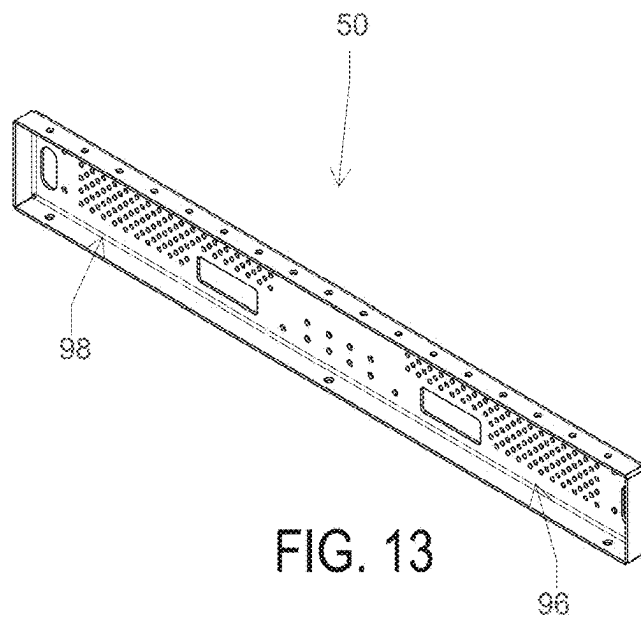


FIG. 12



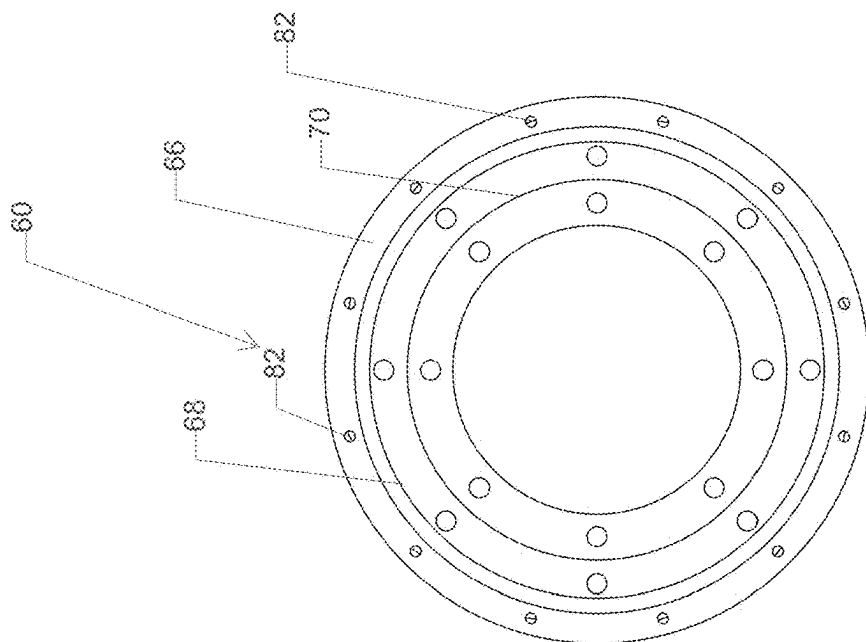


FIG. 16

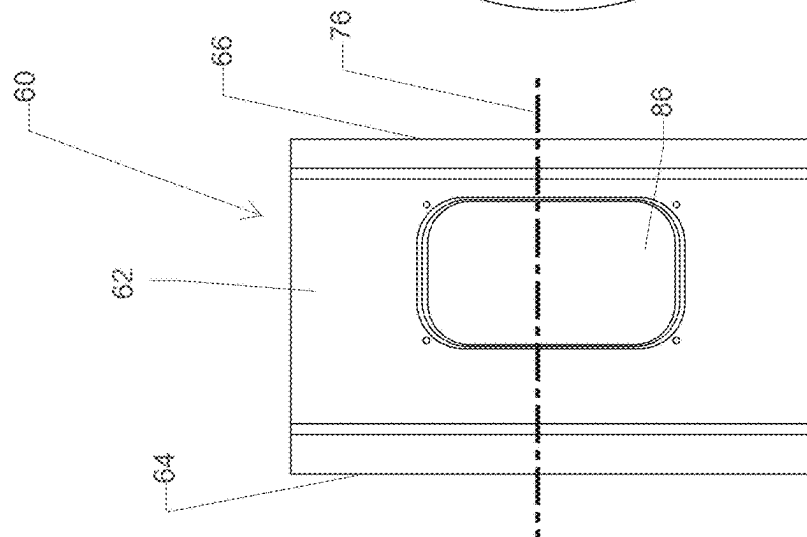


FIG. 17

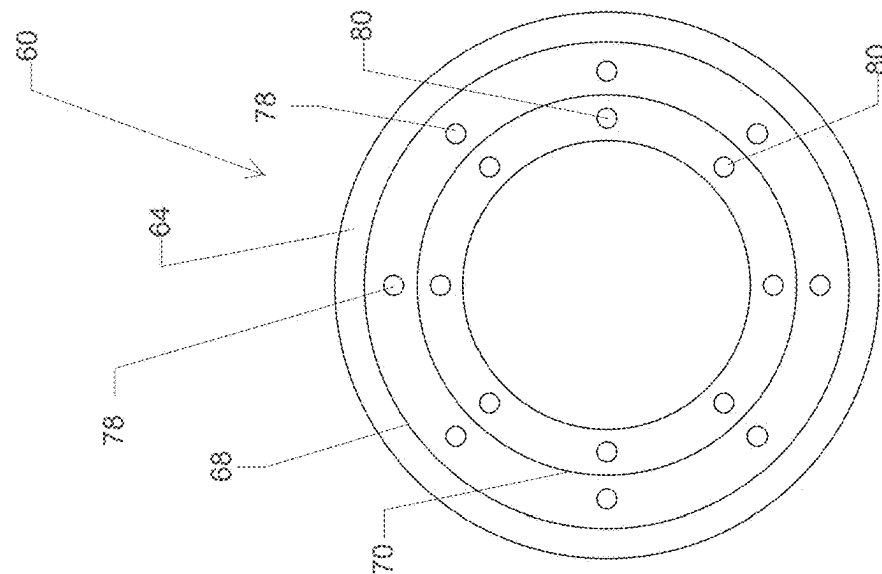


FIG. 18

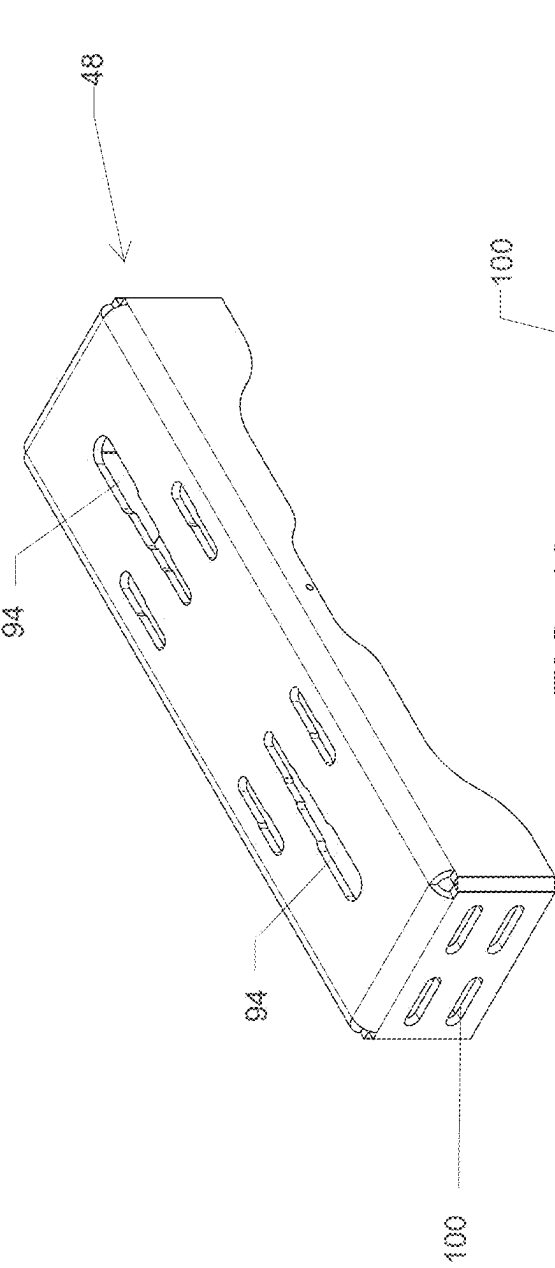


FIG. 19

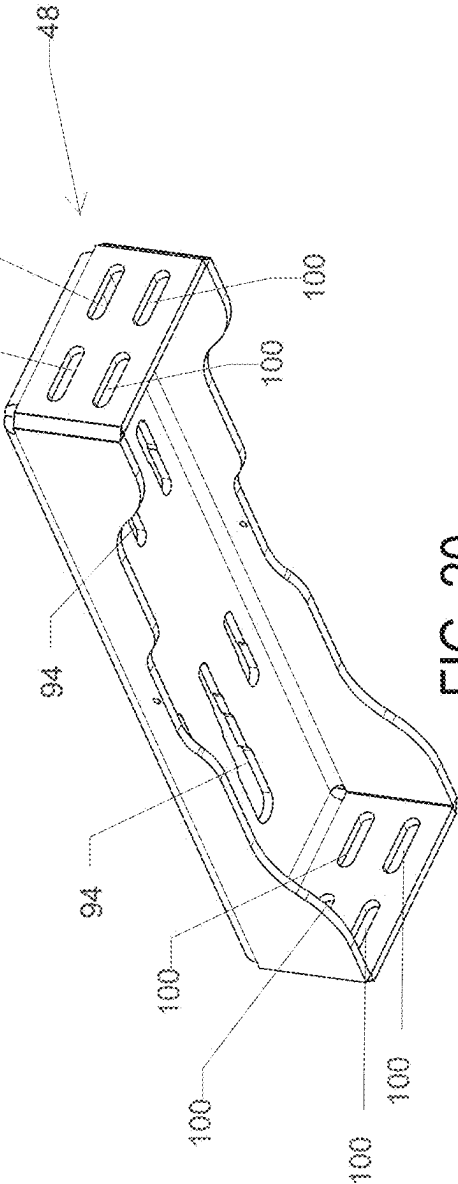
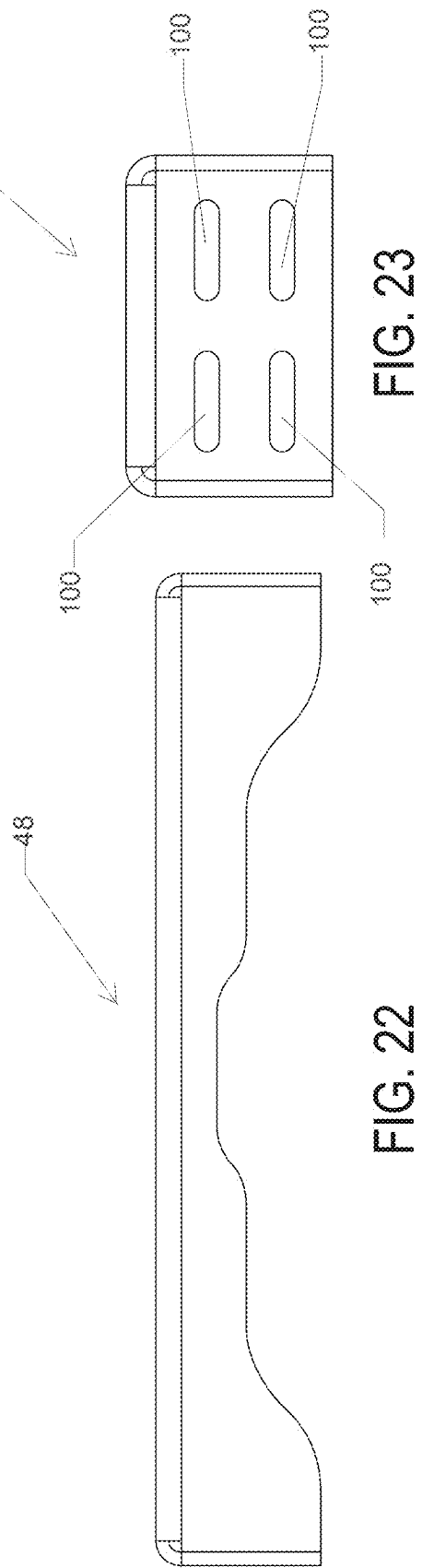
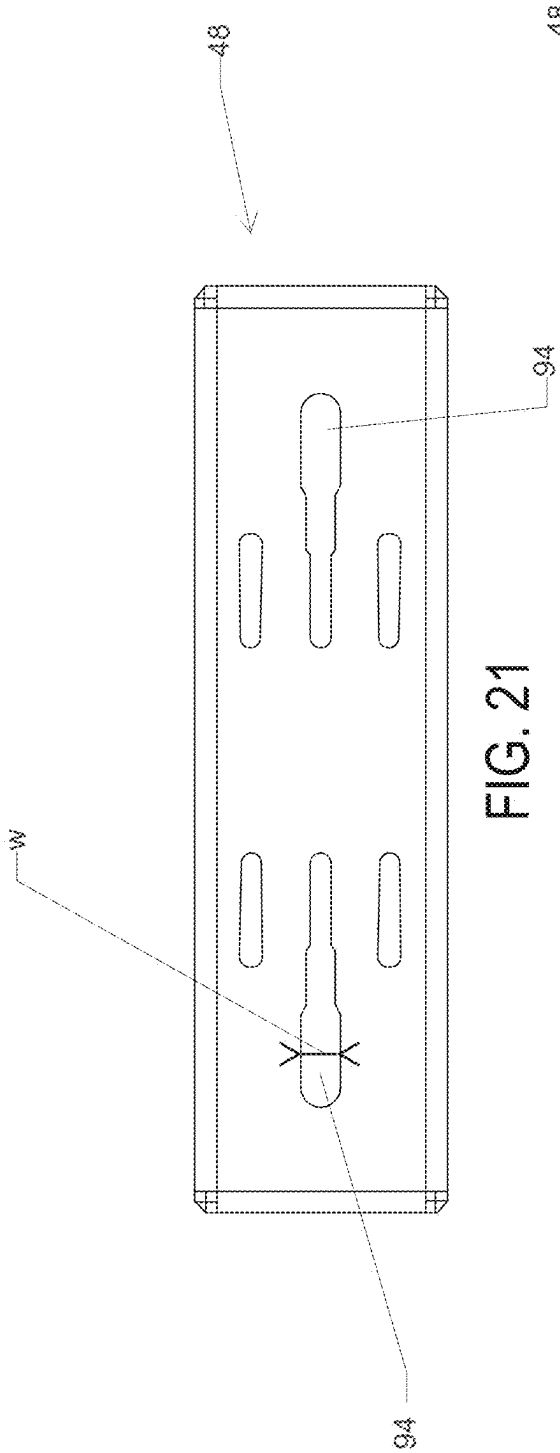


FIG. 20



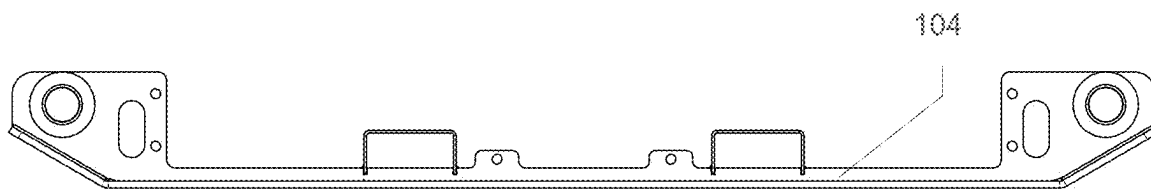


FIG. 24

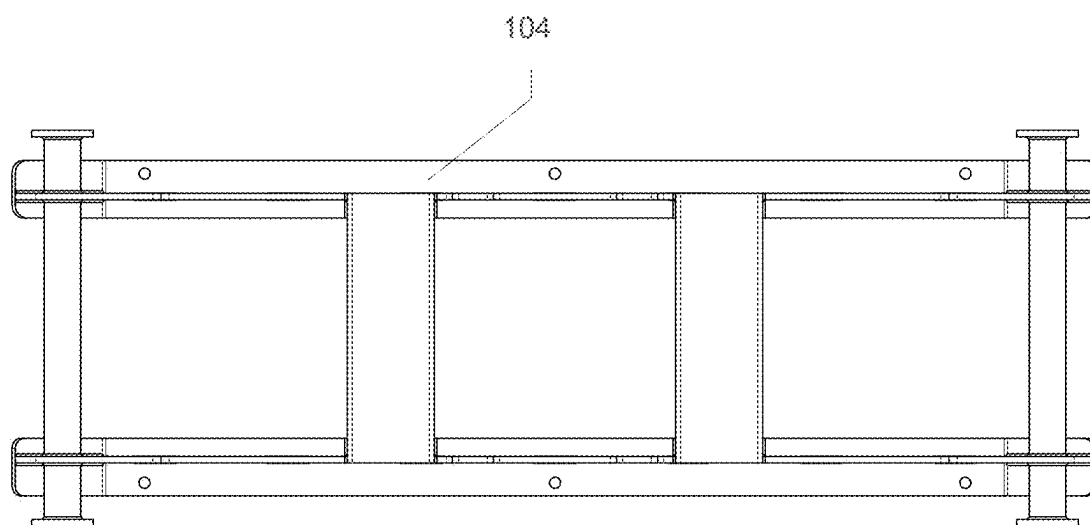


FIG. 25

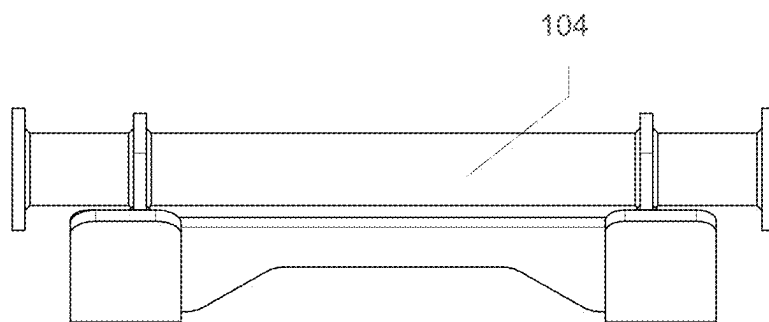


FIG. 26

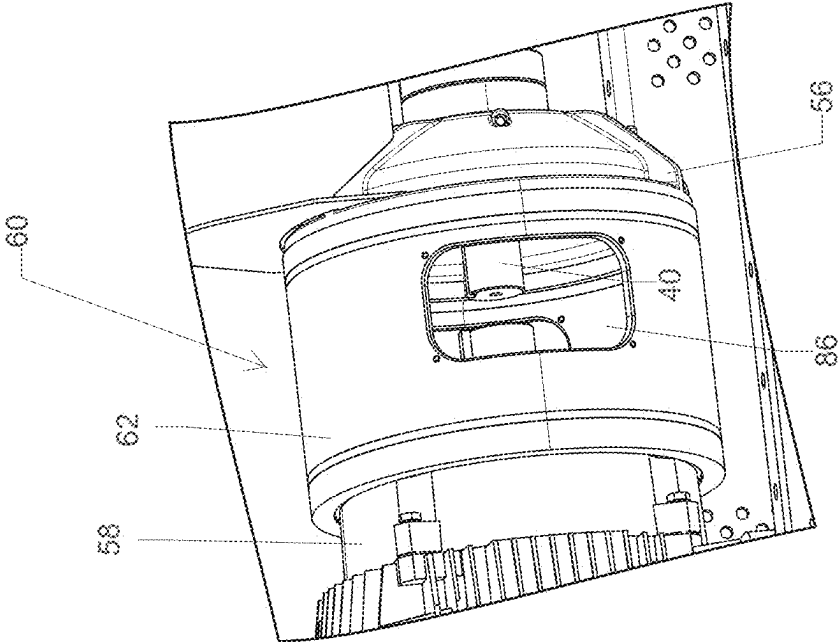


FIG. 27

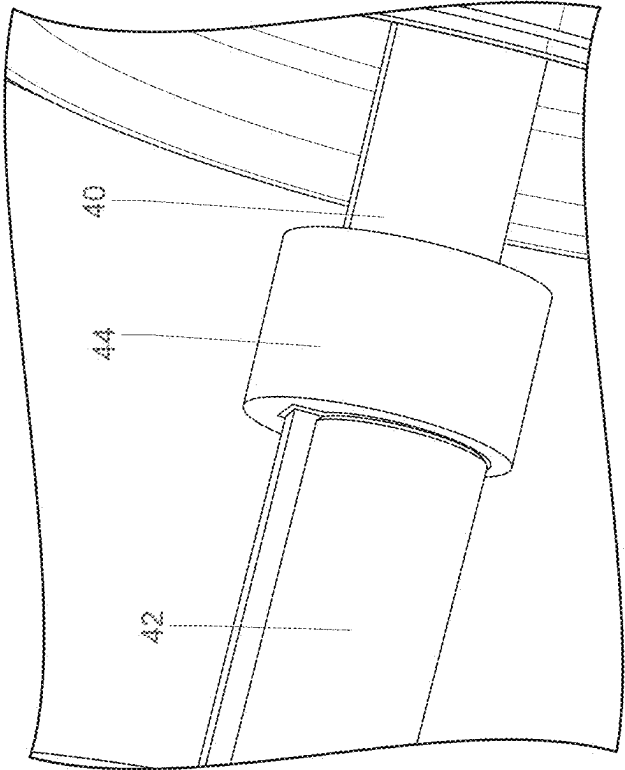


FIG. 28

MODULAR PUMP/MOTOR ASSEMBLY

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application relates to and is a divisional application of U.S. patent application Ser. No. 18/197,986, filed on May 16, 2023, titled MODULAR PUMP/MOTOR ASSEMBLY, the entire contents of which being expressly incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure is directed to a modular and configurable pump and motor subassembly. More particularly, the present disclosure is directed to a system of components that facilitates ease of manufacture of a high number of different pump and motor subassemblies without requiring a similarly high number of component parts.

BACKGROUND

[0003] Pump and motor subassemblies are useful in a variety of implementations, including those requiring that the subassembly be portable. For example, surface dewatering applications are facilitated by portable pump and motor subassemblies. A supporting frame for a pump and motor combination can be rigidly assembled by, e.g., welding custom frame components together to form a support base for the pump and motor combination. Such a bespoke frame is not able to be disassembled without destruction and rigidly supports the motor output shaft in alignment with the pump input shaft. To assemble a high number of pump and motor combinations, bespoke frame components must be custom made to suit the desired assemblies.

SUMMARY

[0004] In one embodiment, the present disclosure provides a modular pump and motor support system, comprising: a motor stool comprising: an annular body including, a first end having a pump index configured to guide securement of the motor stool to a selected pump chosen from a plurality of pumps, and a second end having a first motor index configured to guide securement of the motor stool to a first selected motor chosen from a plurality of motors, and a second motor index configured to guide securement of the motor stool to a second selected motor chosen from the plurality of motors, wherein the first motor index is different from the second motor index; and wherein the annular body defines an access between the first end and the second end, the access sized to allow an operator to operably join a motor shaft to a pump shaft when the motor stool is secured to the selected pump and one of the first selected motor or the second selected motor. One aspect of this embodiment further comprises a plurality of frame side members, each of the plurality of frame side members being nominally identical in size and shape; and a plurality of frame cross members, each of the plurality of frame cross members being nominally identical in size and shape; wherein the plurality of frame side members and the plurality of frame cross members being configured to be assembled using a plurality of fasteners to form a plurality of different frame configurations each adapted to support a unique pump and motor combination. In a variant of this aspect, each of the plurality of frame side members includes a plurality of apertures, each of the plurality of apertures sized to receive

one of the plurality of fasteners. In another variant, each of the plurality of frame cross members defines at least one graduated aperture that defines a progressively larger opening along its length. In a further variant, the at least one graduated aperture of each of the plurality of frame cross members comprises a first graduated aperture spaced from a second graduated aperture. In another variant of this aspect, the plurality of different frame configurations includes a first configuration adapted to operably support a first pump of the plurality of pumps and the first selected motor secured together by the motor stool, a second configuration adapted to operably support the first pump and the second selected motor, a third configuration adapted to operably support a second pump of the plurality of pumps and the first selected motor, and a fourth configuration adapted to operably support the second pump and the second selected motor. In another aspect of this embodiment, the plurality of pumps includes at least a first pump having a first pump characteristic from a pump characteristic class and a second pump having a second pump characteristic from the pump characteristic class, the second pump characteristic different from the first pump characteristic; and the plurality of motors includes at least a first motor having a first motor characteristic from a motor characteristic class and a second motor having a second motor characteristic from the motor characteristic class, the second motor characteristic different from the first motor characteristic. In yet another aspect, the motor stool maintains operative alignment of the pump shaft and the motor shaft, the pump shaft being coupled to the selected pump and the motor shaft being coupled to one of the first selected motor or the second selected motor.

[0005] In another embodiment, the present disclosure provides a system for supporting a selected pump chosen from a plurality of different pumps and a selected motor from a plurality of different motors, the system comprising: a plurality of nominally identical frame side members; a plurality of nominally identical frame cross members; a motor stool; a selected pump chosen from a plurality of different pumps; and a first selected motor chosen from a plurality of different motors; wherein the plurality of frame side members and the plurality of frame cross members are configured to be assembled to form a plurality of frame configurations; and wherein the motor stool has a pump index configured to guide securement of the motor stool to the selected pump, a first motor index configured to guide securement of the motor stool to the first selected motor and a second motor index configured to guide securement of the motor stool to a second selected motor chosen from the plurality of different motors. One aspect of this embodiment further comprises: a lifting bale coupled to a first frame side member of the plurality of frame side members and a second frame side member of the plurality of frame side members; and a drag skid coupled to the first frame side member and the second frame side member. In another aspect, the motor stool includes an annular body, the pump index being positioned at a first end of the annular body and the first motor index and the second motor index being positioned at a second end of the annular body, the first end being opposite the second end. In a variant of this aspect, the annular body defines an access between the first end and the second end, the access sized to allow an operator to operably join a motor shaft to a pump shaft when the motor stool is secured to the selected pump and one of the first selected motor or the second selected motor. In a further variant, the motor stool

maintains operative alignment of the pump shaft and the motor shaft, the pump shaft being coupled to the selected pump and the motor shaft being coupled to one of the first selected motor or the second selected motor. In another aspect of this embodiment, each of the plurality of frame cross members defines at least one graduated aperture that defines a progressively larger opening along its length, the at least one graduated aperture being positioned to align with a first aperture of a first foot of the first selected motor and a second aperture of a second foot of the second selected motor. In another aspect, the plurality of frame configurations includes a first configuration adapted to operably support a first pump of the plurality of pumps chosen as the first selected pump and the first selected motor secured together by the motor stool, a second configuration adapted to operably support the first pump and the second selected motor, a third configuration adapted to operably support a second pump of the plurality of pumps chosen as the first selected pump and the first selected motor, and a fourth configuration adapted to operably support the second pump and the second selected motor. In yet another aspect, the plurality of pumps includes at least a first pump having a first pump characteristic from a pump characteristic class and a second pump having a second pump characteristic from the pump characteristic class, the second pump characteristic different from the first pump characteristic; and the plurality of motors includes at least a first motor having a first motor characteristic from a motor characteristic class and a second motor having a second motor characteristic from the motor characteristic class, the second motor characteristic different from the first motor characteristic.

[0006] In yet another embodiment, the present disclosure provides a system for supporting a pump and a motor, comprising: a motor stool having a first end adapted to couple to the pump and a second end adapted to couple to the motor, the motor stool being configured to receive a pump shaft of the pump and a motor shaft of the motor and having at least one access to allow an operator to operably join the pump shaft to the motor shaft; a plurality of nominally identical frame side members; and a plurality of nominally identical frame cross members; wherein the plurality of frame side members and the plurality of frame cross members are configured to be assembled to form a plurality of different frame configurations for supporting the pump and the motor when the pump and the motor are coupled to the motor stool; and wherein the plurality of different frame configurations includes a first frame configuration formed to support the pump when the pump is a first pump selected from a plurality of different pumps and the motor is a first motor selected from a plurality of different motors, and a second frame configuration formed to support the pump when the pump is a second pump selected from the plurality of different pumps and the motor is a second motor selected from the plurality of different motors. In one aspect of this embodiment, the second end of the motor stool includes a first motor index configured to couple to the first motor and a second motor index configured to couple to the second motor. In another aspect, each of the plurality of frame cross members defines at least one graduated aperture that defines a progressively larger opening along its length, the at least one graduated aperture being positioned to align with a first aperture of a first foot of the first motor and a second aperture of a second foot of the second motor. In yet another aspect, the system further comprises: a lifting bale coupled

to a first frame side member of the plurality of frame side members and a second frame side member of the plurality of frame side members; and a drag skid coupled to the first frame side member and the second frame side member.

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] The above-mentioned and other features and advantages of this invention, and the manner of attaining them, will become more apparent and the invention itself will be better understood by reference to the following description of embodiments of the invention taken in conjunction with the accompanying drawings, wherein:

[0008] FIG. 1 is a perspective view of a modular pump and motor combination supported on a modular frame on a skid; **[0009]** FIG. 2 is another perspective view of the modular pump and motor combination supported on a modular frame on a skid of FIG. 1;

[0010] FIG. 3 is a perspective view similar to FIG. 1, but with the pump and motor removed to better illustrate the motor stool and modular frame;

[0011] FIG. 4 is an exploded, perspective view of the modular frame together with optional frame components, a motor stool, lifting bale, and a skid;

[0012] FIG. 5 is an alternative exploded, perspective view of the components illustrated in FIG. 4;

[0013] FIG. 6 is a top elevational view of an assembly of the components illustrated in FIGS. 4 and 5;

[0014] FIG. 6A is a bottom elevational view of an assembly of the components illustrated in FIGS. 4 and 5;

[0015] FIG. 7 is an end elevational view of an assembly of the components illustrated in FIGS. 4 and 5;

[0016] FIG. 8 is front perspective view of a lifting bale assembly usable with the modular frame of the present disclosure;

[0017] FIG. 9 is a rear perspective view of the lifting bale assembly of FIG. 8;

[0018] FIG. 10 is a rear elevational view of the lifting bale assembly of FIGS. 8 and 9;

[0019] FIG. 11 is a perspective view of a skid on which a modular frame of the present disclosure can be supported;

[0020] FIG. 12 is an alternative perspective view of the skid of FIG. 11;

[0021] FIG. 13 is an exterior perspective view of a frame side member, which is a component of a modular frame of the present disclosure;

[0022] FIG. 14 is an interior perspective view of the frame side member of FIG. 13;

[0023] FIG. 15 is a side elevational view of the frame side member of FIGS. 13 and 14;

[0024] FIG. 16 is a first end elevational view of a motor stool of the present disclosure;

[0025] FIG. 17 is a radial elevational view of the motor stool of FIG. 16;

[0026] FIG. 18 is a second end elevational view of the motor stool of FIGS. 16 and 17;

[0027] FIG. 19 is a top perspective view of a frame cross member, which is a component of a modular frame of the present disclosure;

[0028] FIG. 20 is a bottom perspective view of the frame cross member of FIG. 19;

[0029] FIG. 21 is a plan view of the frame cross member of FIGS. 19 and 20;

[0030] FIG. 22 is a lateral elevational view of the frame cross member of FIGS. 19-21;

[0031] FIG. 23 is an end elevational view of the frame cross member of FIGS. 19-22;

[0032] FIG. 24 is a side elevational view of a skid useable with the modular frame of the present disclosure;

[0033] FIG. 25 is a plan view of the skid of FIG. 24;

[0034] FIG. 26 is an end elevational view of the skid of FIGS. 24 and 25;

[0035] FIG. 27 is a perspective view of a motor stool of the present disclosure, with the guard removed from the access to allow coupling and/or decoupling of the motor output shaft and the pump input shaft; and

[0036] FIG. 28 is a perspective view of a shaft coupler operatively positioned to couple the motor output shaft and the pump input shaft.

[0037] Corresponding reference characters indicate corresponding parts throughout the several views. The exemplification set out herein illustrates an embodiment of the invention, the embodiment disclosed below is not intended to be exhaustive or to be construed as limiting the scope of the invention in any manner.

DETAILED DESCRIPTION

[0038] FIG. 1 illustrates pump and motor combination 30, including pump 32 and motor 34. In the exemplification illustrated, pump 32 is a centrifugal pump having inlet 36 and outlet 38. Pump 32 includes pump input shaft 40 (FIGS. 27 and 28) which is arranged to receive a rotational input to drive an impeller to create a negative pressure at inlet 36 and to force fluid (or a fluid/solids mix) through outlet 38. Motor 34 includes motor output shaft 42 driven by motor 34 when motor 34 is energized. As illustrated in FIG. 28, shaft coupler 44 is utilized to rotationally couple motor output shaft 42 to pump input shaft 40. Use of shaft coupler 44 to couple motor output shaft 42 to pump input shaft 40 will be further described below.

[0039] Throughout this document, any form of the word “couple” includes both arrangements in which the components described as coupled are in direct physical contact as well as arrangements in which the components described as coupled are not in direct contact with each other (e.g., two components are “coupled” via at least a third component, such as pump input shaft 40 and motor output shaft 42 being coupled by shaft coupler 44), but yet still cooperate or interact with each other.

[0040] Pump and motor combination 30 is assembled by choosing pump 32 from a plurality of pumps having different pump characteristics in one or more pump characteristic class such as, e.g., max flow, max head, solids size that can be pumped, efficiency and brake horsepower, and by choosing motor 34 from a plurality of motors having different motor characteristics in one or more motor characteristic class such as, e.g., voltage, speed, torque, and rated horsepower. For example, pump 32 may be chosen from a plurality of pumps including a first pump having a first pump characteristic from a pump characteristic class and a second pump having a second pump characteristic from the pump characteristic class, the second pump characteristic different from the first pump characteristic.

[0041] For example, the first pump may have a flow rate of 1,600 US gpm, max head of 120 feet and be able to pump solids up to 3 inches, while the second pump may have a flow rate of 44,000 US gpm, max head of 126 feet and be able to pump solids up to 5.9 inches. In the example given, the first pump characteristic and the second pump charac-

teristic can be any one of flow rate, max head, and solids size. The plurality of pumps may also include a third pump having a flow rate of 1,400 US gpm, max head of 120 feet and be able to pump solids up to 3 inches. In this example, the pump characteristic of flow rate would be different between the first pump and the third pump, while the pump characteristics of max head and solids size would be the same.

[0042] Exemplary motors of the plurality of motors from which motor 34 is chosen may include a first motor having a peak horsepower of 25 hp @ 3600 rpm, and a second motor having a peak horsepower of 50 hp @ 3600 rpm. Generally, the motor characteristics that differ among individual motors that together make up the plurality of motors from which motor 34 is chosen relate to fuel consumption, continuous horsepower, peak horsepower, and displacement.

[0043] Prior pump and motor combinations were rigidly secured to a rigid frame, e.g., a frame fashioned of frame members welded together. Different pump and motor combinations were possible, but a large stock of rigid frames was needed to accommodate the differing elements of the pump and motor combination. The present disclosure provides a method and apparatus that ameliorates that difficulty.

[0044] Frame 46 of the present disclosure is operatively assembled from a plurality of frame cross members 48 and a plurality of frame side members 50. Each frame cross member 48 is nominally identical to each other frame cross member 48. Similarly, each frame side member 50 is nominally identical to each other frame side member 50. Frame cross members 48 and frame side members 50 form the principal components of a frame 46 for supporting a pump and motor combination thereon. Secondary components such as inlet support 52 and outlet support 54 may also be implemented, but are not needed.

[0045] Frame 46 may be constructed to support a large variety of pump and motor combinations with a stock of only two unique parts, i.e., frame cross member 48 and frame side member 50. To ensure proper operative alignment of motor 34 and pump 32 during pumping operation, motor stool 60 is bolted to motor 34 and pump 32. More particularly, motor stool 60 is bolted to motor housing 58 (FIG. 27) and to pump housing 56.

[0046] Referring to FIGS. 16-18, motor stool 60 includes annular wall 62 spanning motor flange 64 and pump index 66. Motor flange 64 and pump index 66 both extend radially inward from annular wall 62. To account for motors of the plurality of motors from which motor 34 is chosen having differing physical sizes (in addition to different motor characteristics), motor stool 60 includes first motor index 68 and second motor index 70 (see also, e.g., FIG. 5). First motor index 68 is exemplified as a longitudinal wall nominally parallel to longitudinal axis 76 of motor stool 60. First motor index 68 extends a distance of, e.g., one inch and acts as a guide surface to pilot motor 34 into operative engagement with motor stool 60. More particularly, motor housing 58 includes a complementary surface (i.e., a surface that is nominally congruent, but slightly undersized relative to motor index 68) that is aligned with first motor index 68 to index motor stool 60 to motor 34. After piloting motor 34 to motor stool 60, motor securement apertures 78 (only some of which are numbered in the Figs. to avoid unnecessarily complicating the drawings) are aligned with complementary apertures in motor housing 58 to allow fasteners such as lag bolts to secure motor stool 60 to motor housing 58. Align-

ment of motor securement apertures 78 to corresponding apertures in motor housing 58 may also be done together with the piloting step.

[0047] Certain motors of the plurality of motors do not have a surface complementary to first motor index 68, but rather have a surface complementary to second motor index 70. Like first motor index 68, second motor index 70 is exemplified as a longitudinal wall nominally parallel to longitudinal axis 76 of motor stool 60. Second motor index 70 extends a distance of, e.g., one inch and acts as a guide surface to pilot an alternative motor into operative engagement with motor stool 60. More particularly, second motor index 70 acts as a guide for piloting a motor housing having a complementary surface (i.e., a surface that is nominally congruent, but slightly undersized relative to motor index 70) to motor stool 60. After or during the step of piloting a motor chosen from the plurality of motors to motor stool 60 with second motor index 70, motor securement apertures 80 (only some of which are numbered in the Figs. to avoid unnecessarily complicating the drawings) are aligned with complementary apertures in the motor housing to allow fasteners such as lag bolts to secure motor stool 60 to the motor housing.

[0048] Pump index 66 of motor stool 60 is configured to align with any of a variety of different SAE brackets (not shown) that are coupled to the housings of any of a variety of different pumps (e.g., diesel powered pumps) to account for pumps of the plurality of pumps from which pump 32 is chosen having differing physical sizes (in addition to different pump characteristics).

[0049] Pump index 66 is exemplified as a longitudinal wall nominally parallel to longitudinal axis 76 of motor stool 60. Pump index 66 acts as a guide surface to pilot pump 32 into operative engagement with motor stool 60. More particularly, pump housing 56 includes a complementary surface (i.e., a surface that is nominally congruent, but slightly undersized relative to pump index 66) that is aligned with pump index 66 to index motor stool 60 to pump 32. After piloting pump 32 to motor stool 60, pump securement apertures 82 (only some of which are numbered in the Figs. to avoid unnecessarily complicating the drawings) are aligned with complementary apertures in the SAE bracket attached to pump housing 56 to allow fasteners such as lag bolts to secure motor stool 60 to the SAE bracket. Alignment of pump securement apertures 82 to corresponding apertures in the SAE bracket may also be done together with the piloting step.

[0050] With one of pump 32 and motor 34 secured to motor stool 60, the other of pump 32 and motor 34 can be secured to motor stool 60, with the fasteners accessed through access 86. Referring to FIG. 4, motor stool 60 includes two accesses 86, one on either side of motor stool 60. Each access 86 is defined by an opening through annular wall 62. Referring to FIGS. 27 and 28, with motor stool 60 secured to pump 32 and to motor 34, motor output shaft 40, which extends from motor housing 58 into an open interior of motor stool 60 defined by annular wall 62, can be accessed through one of accesses 86. Pump input shaft 40, which extends from pump housing 56 into the open interior of motor stool 60 defined by annular wall 62 can similarly be accessed. Shaft coupler 44 includes a keying feature allowing shaft coupler 44 to be rotationally keyed to both motor output shaft 42 and pump input shaft 40. With pump 32 and motor 34 secured to motor stool 60, as previously

described, motor output shaft 42 is aligned with pump input shaft 40 and shaft coupler 44 can be slid along the rotational axis shared by motor output shaft 42 and pump input shaft 40 from engagement with only one of motor output shaft 42 and pump input shaft 40 to engagement with both so that motor output shaft 42 is fixed for rotation with pump input shaft 40.

[0051] When accesses 86 are no longer needed to allow for securing motor stool 60 in operative position with respect pump 32 and motor 34 and for rotationally securing motor output shaft 42 to pump input shaft 40, guards 88 can be secured by fasteners to annular wall 62 of motor stool 60. In the exemplification illustrated, guards are perforated to allow airflow into and out of the interior of motor stool 60 defined by annular wall 62. Additional motor stools from which motor stool 60 is chosen may also form a part of the kit from which a pump, motor, and frame subassembly may be assembled. More particularly, the present exemplification features a second motor stool having the same design features of motor stool 60, but with the geometry scaled up, for example with a 1.1 scale factor.

[0052] In the exemplification illustrated, two frame cross members 48 are secured to motor 34. Referring to FIGS. 1 and 3, motor 34 includes two sets of feet 90, with a first set of feet 90 extending from motor 34 at a first position along the longitudinal axis of motor 34 corresponding to frame cross member 48a and a second set of feet 90 extending from motor 34 at a second position along the longitudinal axis of motor 34 corresponding to frame cross member 48b. In this document, a “set” or “pair” of feet 90 refers to two feet 90 extending at nominally the same position along the longitudinal axis of motor 34, with each foot of the pair or set extending from motor 34 on an opposite side of a vertical plane including the longitudinal axis of motor 34. Each foot 90 includes at least one aperture 92 through which a fastener such as a lag bolt can be passed to secure foot 90 to a frame cross member 48.

[0053] Referring to FIG. 21, each frame cross member 48 includes a pair of graduated apertures 94 corresponding to a pair of feet 90 extending from motor 34. Graduated apertures 94 each progressively increase in width w from a position closer to the center of the frame cross member to a position further from the center of the frame cross member. Certain motors from the plurality of motors from which motor 34 is chosen have each pair of feet dimensioned such that apertures 92, one from each foot of a pair of feet, are aligned with the inner most, smallest width portion of graduated apertures 94 on a frame cross member 48 to secure these motors to frame cross member 48. Feet of these motors include apertures 92 having a diameter nominally equal to the width of graduated apertures 94 at the smallest width of graduated apertures 94. In this way, the smallest width portion of graduated apertures 94 corresponds to the size of the motor chosen. A fastener is passed through apertures 92 and the smallest width portion of graduated apertures 94 to secure these motors to frame cross member 48.

[0054] Certain other motors from the plurality of motors from which motor 34 is chosen have each pair of feet dimensioned such that apertures 92, one from each foot of a pair of feet, are aligned with the intermediate, middle width portion of graduated apertures 94 on a frame cross member 48 to secure these certain other motors to frame cross member 48. Feet of these motors include apertures 92

having a diameter nominally equal to the width of graduated apertures 94 at the middle (between the largest and the smallest) width of graduated apertures 94. In this way, the middle width portion of graduated apertures 94 corresponds to the size of the motor chosen. A fastener is passed through apertures 92 and the middle width portion of graduated apertures 94 to secure these motors to frame cross member 48.

[0055] Certain alternative motors from the plurality of motors from which motor 34 is chosen have each pair of feet dimensioned such that apertures 92, one from each foot of a pair of feet, are aligned with the outside, largest width portion of graduated apertures 94 on a frame cross member 48 to secure these certain alternative motors to frame cross member 48. Feet of these motors include apertures 92 having a diameter nominally equal to the width of graduated apertures 94 at the largest width of graduated apertures 94. In this way, the largest width portion of graduated apertures 94 corresponds to the size of the motor chosen. A fastener is passed through apertures 92 and the largest width portion of graduated apertures 94 to secure these motors to frame cross member 48.

[0056] Graduated apertures 94 also correspond to a pair of feet extending from pump 32. As with motor 34, certain pumps of the plurality of pumps from which pump 32 is chosen have a pair of feet dimensioned such that an aperture from each one of the pair of feet is aligned with the inner most, smallest width portion of graduated apertures 94 on a frame cross member 48 to secure these pumps to frame cross member 48. Feet of these pumps include apertures having a diameter nominally equal to the width of graduated apertures 94 at the smallest width of graduated apertures 94. In this way, the smallest width portion of graduated apertures 94 corresponds to the size of the pump chosen. A fastener is passed through the apertures through the feet of the chosen pump and through the smallest width portion of graduated apertures 94 to secure these pumps to frame cross member 48.

[0057] Certain other pumps from the plurality of pumps from which pump 32 is chosen have a pair of feet dimensioned such that an aperture from each one of the pair of feet is aligned with the intermediate, middle width portion of graduated apertures 94 on a frame cross member 48 to secure these pumps to frame cross member 48. Feet of these pumps include apertures having a diameter nominally equal to the width of graduated apertures 94 at the middle width of graduated apertures 94. In this way, the middle width portion of graduated apertures 94 corresponds to the size of the pump chosen. A fastener is passed through the apertures through the feet of the chosen pump and through the middle width portion of graduated apertures 94 to secure these pumps to frame cross member 48.

[0058] Certain alternative pumps from the plurality of pumps from which pump 32 is chosen have a pair of feet dimensioned such that an aperture from each one of the pair of feet is aligned with the outside, largest width portion of graduated apertures 94 on a frame cross member 48 to secure these pumps to frame cross member 48. Feet of these pumps include apertures having a diameter nominally equal to the width of graduated apertures 94 at the largest width of graduated apertures 94. In this way, the largest width portion of graduated apertures 94 corresponds to the size of the pump chosen. A fastener is passed through the apertures through the feet of the chosen pump and through the largest

width portion of graduated apertures 94 to secure these pumps to frame cross member 48.

[0059] With pump 32 chosen from the plurality of pumps and motor 34 chosen from the plurality of motors, pump 32 secured to motor 34 via motor stool 60, motor 34 secured to frame cross members 48a, 48b and pump 32 secured to frame cross member 48c, frame 46 can be assembled. Specifically, frame cross members 48a, 48b, and 48c are joined to frame side members 50a, 50b to form frame 46.

[0060] Referring to FIGS. 1-5, and 13-15, each frame side member 50 includes a first array of apertures 96 and a second array of apertures 98. Each of first array of apertures 96 and second array of apertures 98 is formed by a plurality of fastener apertures of nominally identical diameter that are arranged to form an array, i.e., a grouping of fastener apertures having a regular order and arrangement. To assemble frame 46, a plurality fasteners (e.g., lag bolts) having a diameter only slightly smaller (e.g., nominally $\frac{1}{32}$ " smaller) than the fastener apertures of first array of apertures 96 and second array of apertures 98 are each passed through a fastener aperture and into a fastener slot 100 in a frame cross member 48. Fastener slots 100 and the fastener apertures of first array of apertures and second array of apertures are sized and spaced to allow for pump and motor combination 30 to be formed from any one of the plurality of pumps from which pump 32 is chosen and from any one of the plurality of motors from which motor 34 is chosen. While a certain order of construction steps is presented in this exemplary description, this ordering need not be followed.

[0061] Frame side members 50 are all nominally identical. Referring to FIG. 4, frame side member 50a could be rotated 180 degrees about a vertical axis to replace frame side member 50b. Similarly, frame cross members 48 are all identical. Both frame side members 50 and frame cross members 48 are symmetrical about a vertical axis such that $\frac{1}{2}$ of each frame member is a mirror image of the other $\frac{1}{2}$ about a vertical plane containing the axis of symmetry. Therefore, a stock of only two parts, i.e., frame side members 50 and frame cross members 48 is needed to provide frames for all of the possible pump and motor combinations from the plurality of pumps and the plurality of motors from which pump 32 and motor 34 are chosen. Additional supports such as inlet support 52 and outlet support 54 may be secured to frame 46, but are not needed. Inlet support 52 and outlet support 54 can be secured by one or more fasteners such as lag bolts to frame 46 and used to provide support to inlet 36 and outlet 38, respectively.

[0062] Lifting bale 102 may also be secured to frame 46 to provide an attachment point for lifting pump and motor combination 30 supported on frame 46 with, e.g., a crane. Frame 46, with pump and motor combination 30 supported thereon may be supported on drag skid 104 to facilitate transportation. Drag skid 104 includes forklift pockets 106 for receiving the blades of a forklift to facilitate transportation of pump and motor combination 30 supported on frame 46. Forklift pockets 106 align with apertures through frame side members 50 when frame 46 is secured to drag skid 104. Frame 46 may be bolted to drag skid 104 to ensure a secure connection therebetween. For example, frame 46 may include multiple apertures through frame side members 50 outside of first array of apertures 96 and second array

of apertures **98** that align with corresponding apertures through drag skid **104** to receive bolts to secure frame **46** to drag skid **104**.

[0063] While this invention has been described as having exemplary designs, the present invention may be further modified with the spirit and scope of this disclosure. Further, this application is intended to cover such departures from the present disclosure as come within known or customary practice in the art to which this invention pertains.

What is claimed is:

1. A modular pump and motor support system, comprising:

- a motor stool comprising:
 - an annular body including,
 - a first end having a pump index configured to guide securement of the motor stool to a selected pump chosen from a plurality of pumps, and
 - a second end having
 - a first motor index configured to guide securement of the motor stool to a first selected motor chosen from a plurality of motors, and
 - a second motor index configured to guide securement of the motor stool to a second selected motor chosen from the plurality of motors,

wherein the first motor index is different from the second motor index; and

wherein the annular body defines an access between the first end and the second end, the access sized to allow an operator to operably join a motor shaft to a pump shaft when the motor stool is secured to the selected pump and one of the first selected motor or the second selected motor.

2. The modular pump and motor support system of claim 1, further comprising:

- a plurality of frame side members, each of the plurality of frame side members being nominally identical in size and shape; and
- a plurality of frame cross members, each of the plurality of frame cross members being nominally identical in size and shape;

wherein the plurality of frame side members and the plurality frame cross members being configured to be assembled using a plurality of fasteners to form a plurality of different frame configurations each adapted to support a unique pump and motor combination.

3. The modular pump and motor support system of claim 2, wherein each of the plurality of frame side members includes a plurality of apertures, each of the plurality of apertures sized to receive one of the plurality of fasteners.

4. The modular pump and motor support system of claim 2, wherein each of the plurality of frame cross members defines at least one graduated aperture that defines a progressively larger opening along its length.

5. The modular pump and motor system of claim 4, wherein the at least one graduated aperture of each of the plurality of frame cross members comprises a first graduated aperture spaced from a second graduated aperture.

6. The modular pump and motor support system of claim 2, wherein:

- the plurality of different frame configurations includes a first configuration adapted to operably support a first pump of the plurality of pumps and the first selected motor secured together by the motor stool, a second configuration adapted to operably support the first

pump and the second selected motor, a third configuration adapted to operably support a second pump of the plurality of pumps and the first selected motor, and a fourth configuration adapted to operably support the second pump and the second selected motor.

7. The modular pump and motor support system of claim 1, wherein:

the plurality of pumps includes at least a first pump having a first pump characteristic from a pump characteristic class and a second pump having a second pump characteristic from the pump characteristic class, the second pump characteristic different from the first pump characteristic; and

the plurality of motors includes at least a first motor having a first motor characteristic from a motor characteristic class and a second motor having a second motor characteristic from the motor characteristic class, the second motor characteristic different from the first motor characteristic.

8. The modular pump and motor support system of claim 1, wherein the motor stool maintains operative alignment of the pump shaft and the motor shaft, the pump shaft being coupled to the selected pump and the motor shaft being coupled to one of the first selected motor or the second selected motor.

9. A system for supporting a selected pump chosen from a plurality of different pumps and a selected motor from a plurality of different motors, the system comprising:

- a plurality of nominally identical frame side members;
- a plurality of nominally identical frame cross members;
- a motor stool;
- a selected pump chosen from a plurality of different pumps; and
- a first selected motor chosen from a plurality of different motors;

wherein the plurality of frame side members and the plurality of frame cross members are configured to be assembled to form a plurality of frame configurations; and

wherein the motor stool has a pump index configured to guide securement of the motor stool to the selected pump, a first motor index configured to guide securement of the motor stool to the first selected motor and a second motor index configured to guide securement of the motor stool to a second selected motor chosen from the plurality of different motors.

10. The system of claim 9, further comprising:

- a lifting bale coupled to a first frame side member of the plurality of frame side members and a second frame side member of the plurality of frame side members; and
- a drag skid coupled to the first frame side member and the second frame side member.

11. The system of claim 9, wherein the motor stool includes an annular body, the pump index being positioned at a first end of the annular body and the first motor index and the second motor index being positioned at a second end of the annular body, the first end being opposite the second end.

12. The system of claim 11, wherein the annular body defines an access between the first end and the second end, the access sized to allow an operator to operably join a motor

shaft to a pump shaft when the motor stool is secured to the selected pump and one of the first selected motor or the second selected motor.

13. The system of claim **12**, wherein the motor stool maintains operative alignment of the pump shaft and the motor shaft, the pump shaft being coupled to the selected pump and the motor shaft being coupled to one of the first selected motor or the second selected motor.

14. The system of claim **9**, wherein each of the plurality of frame cross members defines at least one graduated aperture that defines a progressively larger opening along its length, the at least one graduated aperture being positioned to align with a first aperture of a first foot of the first selected motor and a second aperture of a second foot of the second selected motor.

15. The system of claim **9**, wherein:

the plurality of frame configurations includes a first configuration adapted to operably support a first pump of the plurality of pumps chosen as the first selected pump and the first selected motor secured together by the motor stool, a second configuration adapted to operably support the first pump and the second selected motor, a third configuration adapted to operably support a second pump of the plurality of pumps chosen as the first selected pump and the first selected motor, and a fourth configuration adapted to operably support the second pump and the second selected motor.

16. The system of claim **9**, wherein:

the plurality of pumps includes at least a first pump having a first pump characteristic from a pump characteristic class and a second pump having a second pump characteristic from the pump characteristic class, the second pump characteristic different from the first pump characteristic; and

the plurality of motors includes at least a first motor having a first motor characteristic from a motor characteristic class and a second motor having a second motor characteristic from the motor characteristic class, the second motor characteristic different from the first motor characteristic.

17. A system for supporting a pump and a motor, comprising:

a motor stool having a first end adapted to couple to the pump and a second end adapted to couple to the motor, the motor stool being configured to receive a pump shaft of the pump and a motor shaft of the motor and having at least one access to allow an operator to operably join the pump shaft to the motor shaft;

a plurality of nominally identical frame side members; and

a plurality of nominally identical frame cross members; wherein the plurality of frame side members and the plurality of frame cross members are configured to be assembled to form a plurality of different frame configurations for supporting the pump and the motor when the pump and the motor are coupled to the motor stool; and

wherein the plurality of different frame configurations includes a first frame configuration formed to support the pump when the pump is a first pump selected from a plurality of different pumps and the motor is a first motor selected from a plurality of different motors, and a second frame configuration formed to support the pump when the pump is a second pump selected from the plurality of different pumps and the motor is a second motor selected from the plurality of different motors.

18. The system of claim **17**, wherein the second end of the motor stool includes a first motor index configured to couple to the first motor and a second motor index configured to couple to the second motor.

19. The system of claim **17**, wherein each of the plurality of frame cross members defines at least one graduated aperture that defines a progressively larger opening along its length, the at least one graduated aperture being positioned to align with a first aperture of a first foot of the first motor and a second aperture of a second foot of the second motor.

20. The system of claim **17**, further comprising:

a lifting bale coupled to a first frame side member of the plurality of frame side members and a second frame side member of the plurality of frame side members; and

a drag skid coupled to the first frame side member and the second frame side member.

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